

UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis Survey
Homework 2: Python practice
Due Wednesday Sep 9, 2:30pm.

Goals: Become sufficiently familiar with Python to:

1. Start Python
2. Allocate, manipulate, and get information about arrays,
3. Print the value of any cell in an array,
4. Change values in an array according to given mathematical or positional criteria (e.g. zero every 5th pixel, set all values greater than 100 to -2000, set a given rectangular sub-array to a given value, insert one array into another),
5. View images and manipulate image colormaps interactively from Python,
6. Make line plots and figures with images, write them in standard formats,
7. Use available resources to find functions, packages, and any instructions you need.

Readings:

1. Do the Matplotlib pyplot tutorial. See <https://matplotlib.org/tutorials/introductory/pyplot.html>.
2. Do sections 1 and 2 of the Using Python for Interactive Data Analysis tutorial. See Files/demos_lectures/week_2/pydatatut/. The exercises are for your edification (don't hand them in).
3. <https://github.com/rougier/numpy-100> . The exercises are for your edification (don't hand them in).

Do not use loops for any problem on this assignment.

Remember to open your `0-ast4762-<yourname>.log` and start logging what you are doing. Also, remember that '`<yourname>`' should be replaced by the username you have chosen.

NOTE: The problems here can be done with much less than the full reading assignment above. Be sure you have mastered the material needed to do the problems, and then get as far as you can in the reading assignment and its exercises. You will be using all of that material, and more, in the rest of the course. This is the only time budgeted for refreshing/learning Python.

NOTE 2: This is the only time that I ask that you **not use AI for these assignments**. Instead, try to do them on your own or with the help of googling/Stackoverflow to get a better feeling/ warm up with Python.

1. **(5 points)** Use your terminal (or PowerShell) to make a new folder `hw2_<yourname>` under your `ast4762/homework` folder. Explain in your log how you did it. For this homework your main homework file is a Python file named `hw2_<yourname>.py` If you prefer to work on a notebook you will start with a `hw2_<yourname>.ipynb` but remember to save it as a `.py` before submission (see end of homework too). Save it in your homework (hw2) folder. Remember to commit your files and push to GitHub as often as needed (also, great backup!).

Your name, assignment number, and the date should appear as comments at the top of the file. At the start of every problem print the problem number using the appropriate Python command. Any remarks or written answers you may make should be printed out using the appropriate Python command.

If you need to comment something in the code (for coding clarity) do so with a normal comment (i.e., `# this is a comment`). Remember that using comments is good coding practice, and it will be taken into account during the grading process.

2. **(10 points)** Without using loops write the necessary Python commands that:

a) a1) Create an array of integers x from 0 to 1000 (up to and including 1000). How many elements do you need?

a2) Print the datatype of the array and the array's minimum and maximum.

Remember to answer all questions raised in the assignment, such as "how many elements do you need?".

b) b1) Re-scale x to contain values from 0 to 2π . ***Don't make a new array***, re-scale the array of question 2a.

b2) Print the minimum and maximum values of the new x array.

c) Make an array y whose values are the sine of the values of x .

d) Print the value of element 234 of y [Note the difference between the 234th element and element 234 in Python!].

3. **(10 points total)** Write the necessary Python commands that:

a) **(5 points)** Plot y vs. x from problem 2c. Make the plot publication-ready using reasonable axis labels etc.

- b) **(5 points)** Save your plot as a PNG using the appropriate Python commands (no screenshots or window dumps from outside Python).

4. **(10 points total)**

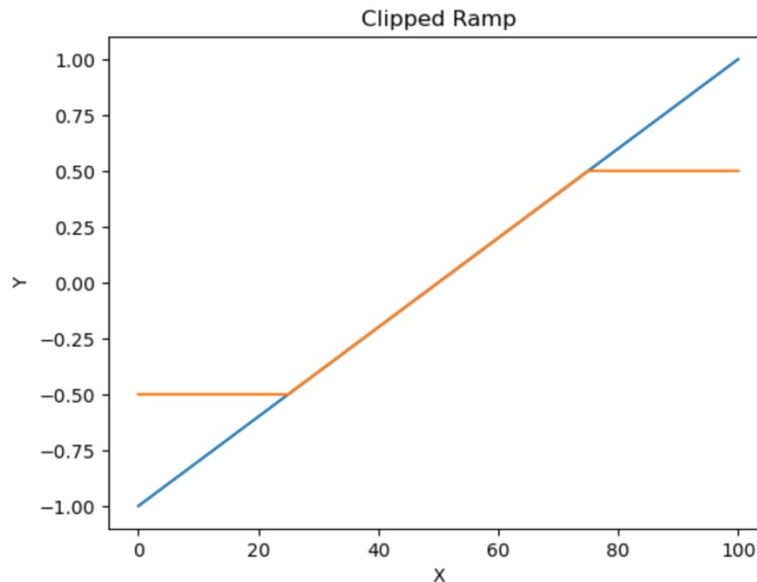
a) **(5 points)**

a1) Make a “ramp” array x with 101 evenly spaced elements going from -1 to +1.

a2) “Clip”, or mask, the array so that any value greater than 0.5 is set to 0.5 and any value less than -0.5 is set to -0.5. There are different ways that you can do this with numpy, but remember, no loops!

b) **(5 points)**

b1) In the same plot, plot the original and clipped arrays. Your figure should look something like this



b2) Use the appropriate python command to save the plot as a PDF. Name the PDF appropriately.

5. **(10 points)** Give the URLs of two web sites outside of UCF that provide free astronomical software that is written in Python. Write a paragraph about each package in your own words. Put the two paragraphs as an extended string (between sets of triple single-quotes) in your main homework file.

6. **(5 points)** Test that your code works in Stokes. If you worked on a Jupyter notebook, make sure to save it as a .py file. Then:

a) Log in to Stokes. Explain how you did this in your log.

- b) Make a folder `hw2_<yourname>`. Explain how you did this in your log.
- c) Copy your `.py` file into that folder. Explain how you did this in your log.
- d) Test it! Does it run without issues (i.e., if you type `python hw2_<yourname>.py` does your code run without crashing) ?
 - a. Yes! Great. Move on to the next steps
 - b. No. Debug!
- e) Don't forget to commit and push to GitHub. Explain what you did to do that in your log.
- f) Make a screenshot that shows your logging history, showing that you committed the file and added it to your `hw2_<yourname>` folder (remember to use an appropriate name for the screenshot! Also, don't forget to copy it to your Stokes folder, too).
- g) If needed, copy the rest of the content of your `homework/hw2_<yourname>` folder into your Stokes folder (e.g., any screenshots, etc., that might have been asked for).
- h) Explain how you will do the following in your log and then copy the log into your Stokes folder:
 - a. Copy the log in the Stokes folder.
 - b. Make a `tar.gz` of the folder in Stokes and explain how you did this in your log.
 - c. Download the `tar.gz` file in your local `handin/` folder.
 - d. Submit the file to Webcourses.
- i) Copy your log into your Stokes folder and make a `tar.gz` file of the `hw2/` folder.

7. (2 points) Submit your homework.